

BAVARIA HIGHWAY TRAFFIC FORECASTING

Forecasting Alpine highway traffic, a year ahead

An automated, data-driven replacement for the hand-built Fahrkalender
— daily congestion forecasts for Bavaria's two busiest Alpine motorway
corridors.



A8 East · A93 South

THE PROBLEM

Replacing a calendar built by hand

Every year, Die Autobahn experts hand-build the **Fahrkalender** — a colour-coded calendar of expected holiday traffic. It is manual, slow to update, and published only months ahead.

Our system learns the same judgement from three years of detector data, then issues it automatically: every day, every corridor, up to a year in advance.

85,094

vehicles on the single busiest day recorded

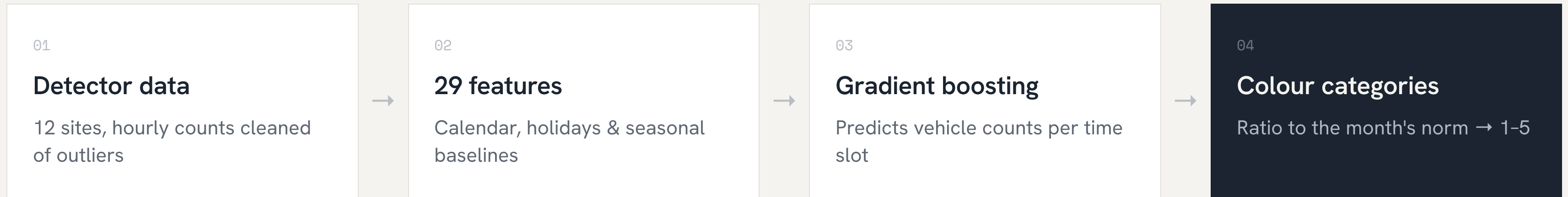
283,160

hourly detector readings, 2023–2025

1 year

forecast horizon, refreshed daily

How the forecast is built



■ Judged against the month, not the year

A busy July day is measured against typical Julys — so a quiet midnight never reads like afternoon rush hour.

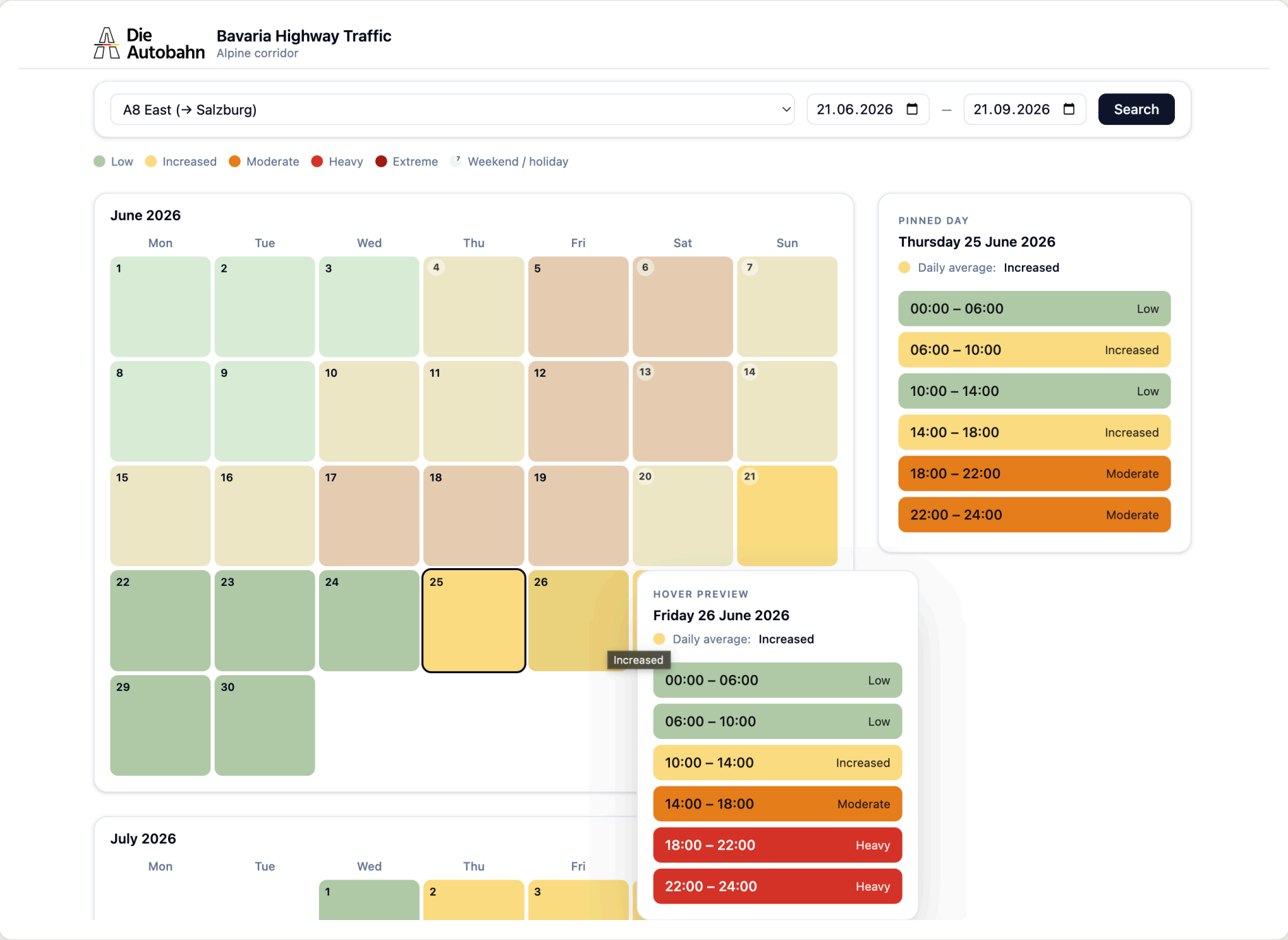
■ Tested on a year it never saw

Trained on 2023-24 and validated on a fully held-out 2025 — no peeking at the future during training.

What every traveller sees

	Low	Free-flowing
	Increased	Slightly above normal
	Moderate	Congestion risk
	Heavy	Expect delays
	Extreme	Significant congestion

Pick a road, direction and date range. Each day takes the colour of its busiest daytime window — with a slot-by-slot breakdown and the reasons behind it on click.



THE RESULTS

How well it predicts

0.923

R^2 — variance explained on
held-out 2025

+35.7%

lower error than a calendar
lookup (MAE)

98.1%

heavy-day detection vs the
official Fahrkalender

73.8%

within one colour of the experts

On the metric that matters most — spotting the
heavy days — the system virtually never misses.